Retrieval-Augmented Generation (RAG) Architecture Overview

Page 1: Introduction to RAG
Retrieval-Augmented Generation (RAG) is a technique that combines an external knowledge base with large lan

Key Advantages:
1. Mitigates hallucination by anchoring generation in verifiable facts.
2. Eliminates the high computational cost of continuously fine-tuning models.
3. Enables private enterprise data to be queried securely without sharing internal data for model retraining.

Page 2: Semantic Chunking and Vector Similarity

Text Chunking Strategy:
Raw text must be divided into smaller passages. LangChain's RecursiveCharacterTextSplitter is often used with

Vector Embeddings:
Chunks are converted into 384-dimensional dense vectors using models like all-MiniLM-L6-v2 from Sentence Tra

FAISS Indexing:
Facebook AI Similarity Search (FAISS) enables millisecond nearest-neighbor lookups using cosine or Euclidean

Page 3: Evaluation Metrics and Future Work

Evaluating RAG Systems:
Two core dimensions are evaluated:
1. Retrieval Relevance: Did FAISS return passages containing the true answer?
2. Faithfulness: Did the LLM answer strictly using retrieved context without hallucinating external claims?

System Limitations:
Standard text extractors cannot read scanned or photographed documents without an OCR engine. In addition, c